

ANTHONY MOLLENKOPF

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EDUCATION

Purdue University

Bachelor of Science, Computer Science

Sep 2024 - May 2028

West Lafayette, Indiana

- Coursework: Data Structures, Linear Algebra, Systems Programming, Algorithms, Software Methodology, Statistics, Computer Architecture, Multivariate Calculus, C Programming, Object Oriented Programming

EXPERIENCE

Sphero Swarm Club

Perceptions Lead

Sep 2025 - Present

West Lafayette, Indiana

- Built a real-time multi-object tracking system using YOLOv8 and OAK-D, supporting 30+ simultaneous objects with unique ID assignment, latency measurement, and a debug GUI.
- Developed an AprilTag-based computer vision pipeline using OpenCV for tag detection, boundary visualization, and homography-based perspective correction to produce a top-down view.

PROJECTS

Vision-Based Rubik's Cube Move Tracker

- Engineered a real-time CV pipeline combining a custom-trained YOLOv8-pose keypoint model, solvePnP pose estimation, and MediaPipe hand tracking to reconstruct full cube state and orientation from webcam input alone.
- Designed a move-detection engine that hypothesis-tests all 18 legal moves against maintained state, using an exact-match/unique-winner rule with time-bounded multi-move search and Kociemba's algorithm as fallback for occluded observations.
- Built a WebSocket-driven three.js frontend rendering the tracked cube state alongside a move log distinguishing directly observed from algorithmically inferred moves.
- Built a custom pygame labeling tool and curated a 1,000+ image keypoint dataset, training via GPU (Colab/Kaggle) for occlusion-robust detection at competitive solving speeds.

FERDA - Wildfire Smoke Detection System

- Fine-tuned YOLOv8 instance segmentation model on custom-labeled smoke dataset (3 classes: Smoke, Ground, Sky), achieving 0.90 mAP50 (box) and 0.95 mAP50 (mask) on held-out validation data.
- Engineered post-processing pipeline with color-coded segmentation masks and geometric smoke-base extrapolation to project estimated fire locations from 2D aerial imagery.
- Built multithreaded desktop GUI for real-time inference with adjustable confidence thresholds, zoom/pan controls, and binary mask visualization; developed companion OpenCV color-filter tool for video preprocessing and export.
- Evaluated thermal vs. multispectral imaging tradeoffs for drone-based detection and proposed multi-source training strategy (HPWREN, FLAME, Boreal Forest UAV datasets) to address cross-domain generalization limits.

StageSense - AI Speech Coaching App

- Collaborated on a 4-person team to build an AI-powered speech coaching application that provides real-time feedback on delivery, pacing, and filler word usage.
- Developed frontend components and integrated speech-to-text processing to analyze user recordings and generate actionable coaching insights.
- Followed Agile methodology with sprint planning, code reviews, and CI/CD workflows using Git and GitHub.

TECHNICAL SKILLS

Languages: Python, Java, C, C++, JavaScript, R, SQL

Developer Tools: VS Code, Cursor, Git, Linux, Terminal

Technologies/Frameworks: React, OpenCV, PyTorch, YOLOv8, MediaPipe, three.js, WebSockets, VexFlow, Web MIDI API, Tkinter, JUnit, GitHub

EXTRACURRICULAR

Victoria National

Groundskeeper

2024 - 2025

- Maintained course quality at a top-50 U.S. golf course, performing mowing, irrigation, bunker raking, and treatment application to ensure optimal playing conditions.